robust PPI intervention to improve recruitment and/or retention in surgical trials.
Methods: The study comprises four stages: (1) Mapping current PPI practice in UK surgical trials; (2) Focus groups with stakeholders (surgical trial investigators, administrators and patient or lay contributors) to explore their views on PPI, recruitment and retention; (3) Survey of stakeholders' views about possible components of a PPI intervention; (4) Consensus workshop with selected stakeholders to design a PPI intervention for evaluation.
Preliminary results: Active, UK-led, adult surgical trials were eligible for the mapping survey; 129 were identified and 72 (59%) were included. Sixty-six (92%) reported PPI in designing the trial (79%), undertaking the trial (26%), analysing or interpreting results (31%) and/or disseminating findings (60%). Patients and/or lay people were usually involved as consultants (63%) and/or members of the trial steering committee (72%), and sometimes as grant co-applicants (36%) and/or members of the trial management group (35%). Developing participant information materials was the single most common PPI activity, reported in 71% of included trials.
Conclusions: UK surgical trials involve patients and members of the public in a variety of different ways, most commonly at the beginning and end of the trial lifecycle and in oversight or advisory roles. This knowledge will inform the development of a robust PPI intervention aimed at improving recruitment and retention in surgical trials.

0022: OSSEOINTEGRATED JOINT REPLACEMENT CONNECTED TO A LOWER LIMB PROSTHESIS: AN IDEAL STAGE 1 PROOF OF CONCEPT STUDY WITH 7 CASES

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Background: One third of lower-limb-amputees experience socket-residuum interface problems, leading to reduced prosthetic use and quality-of-life. Bone-anchored-implants have been developed to eliminate this interface. Our findings from a prospective study suggest favorable benefit-harm trade-off with improved mobility and 100% 2-year implant survival with no major complications(including deep infection). However, this technique is contraindicated in amputees with short residuums' due to small bone-implant-contact. We aim to report on feasibility and safety of combining total joint (hip/knee) replacement (THR/TKR) with osseointegrated implants for prosthetic attachment in this patient group.
Methods: We reviewed all patients who underwent this procedure in 2012-2014 at our centre. In a 2-stage procedure, a custom-made-implant was connected to THR/TKR and a stoma was created. Validated measures of health-related quality-of-life and ambulation were assessed at baseline and follow-up(24-44months) including the Questionnaire-for-Transfemoral-Amputees(Q-TFA) and 6-minute-walk-test(6MWT). All adverse events were classified as major if they required surgical management.
Results: Seven(THR-3/TKR-4) patients aged 35-77(46) years underwent the procedure. Two cases of superficial infection occurred but resolved with antibiotics without major complications. All patients showed improved Q-TFA and SF-36 scores (43 and 22 respectively). All 3 wheelchair-bound patients became community ambulators; the four other patients exhibited improved ambulation.
Interpretation: This proof-of-concept case-series presents the 1st report to combine osseointegration and THR/TKR. Meticulous planning, systematic recording of all outcomes indicate that this procedure is feasible for this challenging group. A prospective database has been developed by us to collect long-term data, better estimate safety and benefits, inform ongoing technical improvements and refine patient selection criteria.

0023: THE EAST GRINSTEAD CONSENT COLLABORATIVE (EGCC) ONLINE CONSENT CREATOR

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Background: Litigation surrounding matters of consent is increasingly seen, the NHS Litigation Authority paid out £7million for compensation regarding inadequate consent in 2010. The East Grinstead head and neck consent forms were introduced three years ago and were well received by both patients and surgeons. We wanted to offer the same protection to our colleagues, but firstly, we needed to determine nationwide practices of consent. The East Grinstead Consent Collaborative (EGCC), the first maxillofacial trainee collaborative project, was formed to determine this. Using the results submitted by oral and maxillofacial (OMF) units nationwide; the project was taken to the next stage by developing an online patient specific consent form creator to allow surgeons to create patient tailored consent forms. We want to be able to offer patients and clinicians greater transparency prior to undergoing major head and neck surgery.
Results: We have received data for 1200 patients from 15 units and data collection will be complete by January. Preliminary analysis has shown significant variability in content ($p<0.001$, 1-way ANOVA) with only 27% warned of the risk of death for major head and neck ablative surgery and reconstruction. We are currently in the development stage of this online tool and are anticipating collaborating with the British Association of Oral and Maxillofacial Surgeons (BAOMS) to host this tool upon successful application for funding.

0024: DEVELOPMENT OF AN ONLINE PLATFORM FOR REGISTRATION AND OUTCOME MEASUREMENT OF UROGYNECOLOGICAL IMPLANTS ACCORDING TO IDEAL-SYSTEM

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Background: Most aspects of implants for reconstruction of the pelvic floor are still under debate or poorly studied. Different tools and definitions of success make the comparability of studies and meta-analyses in this field of surgery difficult. The aim of the present study was to prove the feasibility of a register to analyse the outcome after implant application.
Methods: The available literature on the application of implants for pelvic organ prolapse (POP) and female and male stress urinary incontinence (SUI) repair was reviewed according to IDEAL stages of surgical innovation. A working group was formed to create an online platform for registration and outcome measurement of implant-assisted operations for POP and SUI repair. To date, 20 patients from a previous published study on modified mesh materials were evaluated over 23 months follow up in the register. For validation a previously published modified "satisfaction, anatomy, continence, safety – S.(A.)C.S score" was used.
Preliminary results: A review of the literature revealed missing data especially on male continence surgery and long-term trials. A register was established with consensus on clear definitions and classifications of patient variables, surgical procedures and mesh materials used, as well as outcome parameters. According to the S.(A.)C.S. scoring system, only 14 patients (70 %) reached the maximum score of cure.

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[+] Members of the IDEAL Collaboration, both authors contributed equally to this manuscript.

Conclusion: An online platform for registration and outcome measurement of implant-assisted POP and SUI repair with clear definitions and classifications is offered to the surgical community.

0026: BRINGING LAPAROSCOPY TO YOUR OWN HOME

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Background: Surgical procedures are now opting for a minimally invasive procedure versus the traditional open procedure. The legal maximum limit on the average number of working hours is currently set at 48 per week within the United Kingdom, a reduction in training hours has had a considerable negative effect on surgical training. Doctors are working longer hours voluntarily to gain the skills they need for laparoscopic surgery, which consists of a longer learning curve and costly simulators. Our alternative is a home made boxed laparoscopic trainer, which is low cost, available for all trainees and help them train at their own convenience.
Material and methods: We designed an inexpensive laparoscopic training device composed of a webcam and LED lights mounted in a specifically designed wooden box, the total cost of which was 119 British pounds. Practice exercises using a geoboard and poultry were performed.
Preliminary results: Our device is an inexpensive trainer, which anyone may build and use in his or her own home. The aim of the trainer is not to simulate real life operations in a live individual but to aid the individual to practice motor co-ordination, handling of laparoscopic instruments and achieve spatial orientation in a two dimensional environment prior to theatre.

0030: AUTOLOGOUS PLASMA COATING IMPROVES THE BIOCOMPATIBILITY OF MESH IMPLANTS. ON THE IDEAL WAY FROM BENCH TO BEDSIDE

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Background: Several FDA warnings relating to severe side-effects led to discussions concerning the biocompatibility requirements of surgical meshes. Our aim was to develop a standardized and manufacturer-independent in vitro test system for the adherence performance of tissue clusters (fibroblasts, endothelial cells and muscle-derived cells) as a marker for the biocompatibility of commercially available meshes. In addition we investigated different coating strategies for meshes.
Material and methods: The entire experimental approach followed the recently developed recommendations of IDEAL (Innovation, Development Exploration, Assessment and Long-term) for surgical innovations. We developed an in vitro test systems investigating the tissue ingrowth score on randomised meshes as predictive for their biocompatibility. As a development of this approach we investigated different coating strategies of the meshes and the influence of the different coatings on the in vitro biocompatibility. In order to explore the predictive value and validity of the test system and also newly tested coating strategies, we translated the preliminary in vitro results into in vivo circumstances and conducted a large-animal experiment in sheep. In a next step we applied our mesh modification strategy in a human setting.
Results: In the in vitro test system, we could establish a repeatable ranking of meshes with regard to their biocompatibility. The in vitro test system has been shown to be a feasible pattern for the investigation of different mesh coating strategies. Coating of meshes prior to cultivation, with blood plasma, increased the ingrowth score, suggesting improved biocompatibility in vitro by this strategy. The previous ranking of native meshes remained consistent after coating. In a subsequent long-term animal study, we demonstrated that the recently developed in vitro test system predicts the in vivo performances of the meshes. The coating of meshes with autologous plasma prior to implantation had positive effects on the biocompatibility of meshes in vivo. Investigation of the ultra-short-term determinants of the foreign body reaction (FBR) at the implant site in vivo revealed that the local inflammatory reaction is an early and susceptible event after mesh implantation. In a first conducted observational study in human, the procedure of mesh coating with autologous plasma has been successfully applied and been shown to be a safe and feasible method.
Conclusion: The project is continuing. A prospective randomized trial is planned, comparing naive meshes to plasma coated meshes with regard to the biocompatibility and morbidity outcome. As final step of the project we are planning to develop a long term registry to report clinical outcome features of this new method in the future.

0033: PERCUTANEOUS NEPHROLITHOTOMY (PCNL) UNDER LOCAL INFILTRATIVE ANESTHESIA WITH AND WITHOUT STAND-BY ANESTHESIA – COMPLICATION RATES AND CLINICAL OUTCOME FOR A METHOD AT THE EXPLORATION STAGE ACCORDING TO IDEAL

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Background: This retrospective study aims to compare feasibility, safety and complication rates of percutaneous nephrolithotomy (PCNL) under local infiltration anesthesia alone (Group I) and additive intravenous analgetics and / or sedative medications (Group II).
Material and methods: 439 patients have been included in the study, 226 in Group I, and 213 in Group II. The mean American Society of Anesthesiologists score (ASA) was 2.15 ±0.37 (range, 1-4). Demographic characteristics and stone characteristics have been evaluated to determine feasibility, complication rates for safety, and stone-free rates for effectiveness. The study has been retrospectively classified to the IDEAL stages of surgical innovation.
Results: All patients who accepted local infiltration anesthesia underwent PCNL successfully. Of the 439 patients, 138 had pelvic calculi, 173 renal calculi, 66 partial staghorn, 48 complete staghorn, and 16 upper ureteral stone. The total stone free rate in patients was 78.4% over all stone localizations. Compared to the possibility of using additive intravenous analgetics and / or sedative medications we could show differences in the median age, number of tracts, operation duration, hemoglobin drop, fever, transfusion rate, and stone free rate, but not for severe complications such as perirenal hematoma, colon perforation, pleura perforation, AV fistula, skin fistula, and mortality rate.
Conclusion: PCNL performed under local infiltration anesthesia is feasible and provides satisfactory positive clinical outcomes. In this retrospective analysis both kinds of treatment show similar in success and complication rates. The method can be assigned to the E level according to IDEAL.

0034: THE LANDSCAPE OF SURGICAL INNOVATION IN ROBOTIC MICROSURGERY

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Introduction: We systematically assessed the design of studies on robotic microsurgery in their relationship to the IDEAL stages of surgical innovation.